CS-104 Quiz 3

Fall 2026

Name: _____

This is an in-class, closed-book quiz (30 points). If you have any questions, please feel free to ask them. When you are finished, you can turn it in and go.

1. (6 points)

Write a function called `average_above` that receives a list of numbers `values` and a number `limit`, and returns the average of only the elements that are greater than `limit`. If no element is greater than `limit`, it returns `0`. For example, `average_above([3, 8, 1, 9, 5], 4)` returns `7.333...` (the average of 8, 9, and 5). Then write one line that calls your function with that same example and prints the result.

2. (3 points)

The code below crashes. What is printed before the crash, what error happens and why, and how would you fix the function?

```
def discount(price):
    new_price = price * 0.9
    print(new_price)

# --- main program ---
final = discount(50) - 2
print(final)
```

3. (4 points)

Write a function called `cone_volume` that receives the radius (`radius`) and height (`height`) of a cone, and returns its volume. This can be computed as:

volume = (1/3) × pi × radius² × height

where `math.pi` gives the value of pi (assume `import math` has already been done). Be sure to use appropriate Python syntax. (You do not need to provide an algorithm, documentation, or test cases.)

4. (3 points)

Using your `cone_volume` function from the previous question, write a code segment that computes the **total** volume of two cones, one with radius 3 and height 4, and one with radius 1 and height 9, and prints the total rounded to 1 decimal place.

5. (3 points)

Give the array each expression creates (1 point each). Assume `import numpy as np`.

```
>>> np.arange(4)
```
[0 1 2 3] — example answer

```
>>> np.arange(2, 12, 4)
```

```
>>> np.linspace(0, 2, 5)
```

```
>>> np.arange(0, 1, 0.3)
```

6. (4 points)

A cup of coffee cools down in a room. Its temperature after `t` minutes is

T = 20 + 70 × e^(-0.1 t)

Write a code segment that uses NumPy to create 300 evenly spaced times from 0 to 60 minutes, computes the temperature at each time (use `np.exp`), and uses Matplotlib to plot temperature versus time with the axis labels `Time (min)` and `Temperature (C)` and a title.

7. (4 points)

What does this code print? Write every line of output.

```
def puzzle(p, q):
    print("check", q)
    if p < q:
        return q - p
    return p * 3

# --- main program ---
x = puzzle(2, 7)
y = puzzle(4, 1)
z = puzzle(q=5, p=5)
print(x, y, z)
```

8. (3 points)

What does each `print` statement output (1 point each)? In one sentence, explain why the first result is so different from the second.

```
import numpy as np
values = [10, 20]
arr = np.array(values)
print(values * 2)
print(arr * 2)
print(arr + 5)
```